

Reproducing the Dawes 2022/2023 Autocorrelation Search for Multiply Lensed and Binary Quasars in the DESI Legacy Surveys

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Internal Reproduction Report

Abstract

We reproduce the algorithmic core of the [Dawes et al. \(2023\)](#) “autocorrelation” search for multiply lensed and binary quasars in the DESI Legacy Imaging Surveys. The published autocorrelation step is a friends-of-friends (FoF) angular grouping of quasar targets at a fixed linking length — searched first at $10''$, then halved to $<5''$ — followed by Gaia EDR3 proper-motion and parallax cuts and human visual grading. Because Dawes’ $\sim 5\text{M}$ -target *photometric* DESI Quasar Sample is not on disk, we run the identical `spherimatch` FoF ([Hsu, 2024](#)) on the most directly available proxy — the 1,645,843-spectrum DESI DR1 *spectroscopic* QSO sample from `zall-pix-iron.fits`. The proxy is $\sim 1/3$ the size and is selected differently, so absolute candidate counts cannot match the published 436; what we reproduce faithfully is the *algorithm* and the *recovery of discoverable systems*. Our headline result: every published candidate that is *discoverable in our proxy* (has ≥ 2 distinct DR1 spectroscopic QSOs within $5''$) is correctly grouped by our FoF — $58/58 = 100\%$, the apples-to-apples analog of the paper’s $94/94 = 100\%$ recovery of its “discoverable known” systems. Recovered matches sit a median $0.139''$ from the published centroid. The full 1.6M-QSO FoF runs in $\sim 3\text{s}$. Visual-inspection grading (Grades A/B/C) is explicitly out of scope, exactly as in the companion [Hsu et al. \(2025\)](#) reproduction.

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1 Introduction

This report documents the reproduction of Dawes et al. (2023) (ApJS 269:61; arXiv submitted 2022, hence the “Dawes 2022/2023” label), one of a series of internal reproductions covering Prof. Xiaosheng Huang’s strong-gravitational-lensing program. Companion reports cover Huang et al. (2020) (DECaLS ResNet lens finder) and Hsu et al. (2025) (pair-wise spectroscopic lens search over DESI DR1); the present work is the quasar-lens analog of the latter.

What the paper did. Multiply lensed quasars are powerful cosmological probes: the time delay between their multiple images yields an independent measurement of H_0 (Dawes et al., 2023; Treu and Marshall, 2016). To grow the small known sample, Dawes et al. (2023) search the 19,000 deg² DESI Legacy Imaging Surveys (Dey et al., 2019) for new lensed and binary quasars. They apply an *autocorrelation algorithm* to the ~ 5 million objects classified as quasars in the DESI Quasar Sample, visually inspect and rank the resulting systems, and present **436 new multiply lensed and binary quasar candidates**.

The autocorrelation algorithm (§3). The paper’s “autocorrelation” step groups quasar targets that lie within an angular separation cut and “recursively connects multiple pairs of quasar targets within the same candidate system (such as would occur in ‘triples’ and ‘quads’)”. This is precisely a **friends-of-friends** grouping with a fixed angular linking length. Dawes first search a 10'' radius (“to err on the side of completeness”), then halve the cut to $< 5''$ because $> 95\%$ of the discoverable known systems have images separated by $< 5''$. They test the algorithm on the 94 “discoverable known” systems and on simulations, **achieving a 100 % success rate**, then deploy it on the full ~ 5 M quasar-target sample.

Lensed-quasar system bookkeeping (Table 1). Of the ~ 300 known lensed-quasar systems in the footprint, 94 are “discoverable known” (have ≥ 2 objects in the DESI Quasar Sample so the algorithm *can* group them), of which 72 are “discoverable confirmed”. The new output is the 436 candidates.

Post-FoF stellar rejection (§4.1). Milky-Way stellar contaminants are rejected using Gaia EDR3 (Gaia Collaboration et al., 2021) proper motion and parallax. Dawes define significances $\text{PMSIG} = \sqrt{\mu_\alpha^2 + \mu_\delta^2} / \sqrt{\sigma_{\mu_\alpha}^2 + \sigma_{\mu_\delta}^2}$ and $\text{PXSIG} = \varpi / \sigma_\varpi$, and accept a source as extragalactic if $\text{PMSIG} < 8$ and $\text{PXSIG} < 3.5$. Gaia information is available for at least one image of $380/436 \approx 87\%$ of the published candidates.

Scope. As with the Hsu et al. (2025) reproduction, the published pipeline has an algorithmic stage (autocorrelation FoF + Gaia cuts) followed by a human visual-inspection / grading stage (Grades A/B/C). **This reproduction covers the algorithmic stage only.** Visual grading is not attempted.

2 The proxy sample (and why absolute counts cannot match)

The single most important caveat of this reproduction is the input catalog. Dawes ran on the DESI Legacy Imaging Surveys *photometric* DESI Quasar Sample — ~ 5 million Random-Forest-selected QSO *targets*, one detection per source. **That photometric target catalog is not on disk.** We therefore substitute the most directly available proxy: the DESI DR1 *spectroscopic* QSO sample carved from the shared 28.4M-fiber `zall-pix-iron.fits` (DESI Collaboration et al., 2025) via the pre-filter cascade in Table 1.

Table 1: QSO pre-filter cascade on `zall-pix-iron.fits` (DR1).

filter	remaining
raw rows	28,425,963
ZCAT_PRIMARY == True	27,547,223
ZWARN == 0	20,415,199
SPECTYPE == QSO	1,645,843
Dawes input (photometric, NOT used)	$\sim 5,000,000$

This proxy is a related but **not identical** sample: it is $\sim 1/3$ the size and selected differently (spectroscopically confirmed QSOs only, vs. photometric RF-selected targets). Two consequences follow and are quantified honestly throughout this report:

1. **Absolute candidate counts cannot match** 436. Most published candidate *images* are faint, blended lensed images that DESI never spectroscopically confirmed as QSOs. Only 138/436 published candidates have *any* DR1 QSO within $5''$, and only 58/436 have ≥ 2 *distinct* DR1 QSOs (i.e. are “discoverable” in our proxy).
2. **The reproduction targets are the algorithm and the recovery of discoverable systems**, not the raw 436. We assess these in §5.

3 Pipeline

The pipeline is implemented in seven numbered scripts at `reproductions/dawes-2022/`, summarized in Table 2. The FoF engine is the exact `spherimatch` (Hsu, 2024) tool used by the Hsu et al. (2025) pipeline; the catalog fetch and Gaia queries use `astroquery` (Ginsburg et al., 2019); sky-coordinate matching uses `astropy` (Astropy Collaboration, 2022).

FoF grouping. For each of the two linking lengths ($\ell = 5''$ and $10''$) we call `spherimatch.fof` on the (`TARGET_RA`, `TARGET_DEC`) of the 1,645,843 QSOs and retain groups of size ≥ 2 . The $5''$ run takes 2.8 s, the $10''$ run 2.8 s; loading the 28.4M-row catalog dominates the wall-clock at 30 s.

Refinement. The raw FoF groups are dominated by pairs at $\sim 0''$ separation: distinct `TARGETIDS` at the *same* sky position (the same physical QSO observed under multiple DESI surveys/programs, or Tractor splits). Dawes’ photometric target catalog has one detection per source and so never sees this pile-up. Script 05 collapses near-zero-separation duplicates (dedup within $0.2''$) and keeps only “resolved” systems with maximum internal image separation in $[0.3'', 5'']$ — $0.3''$ matches the minimum published Dawes separation of $0.32''$.

Table 2: Reproduction scripts.

script	purpose
00_fetch_published_catalog.py	download Dawes Table 2 from VizieR J/ApJS/269/61 (875 image-rows $\rightarrow$ 436 systems).
01_build_qso_sample_and_fof.py	build the DR1 QSO proxy sample; <code>spherimatch</code> FoF at 5'' and 10''.
02_recover_published_candidates.py	nearest-group recovery of the 436 published positions.
03_gaia_pm_px_cuts.py	Gaia EDR3 PMSIG/PXSIG stellar rejection (checkpointed/resumable).
04_make_figures.py	separation histogram + recovery-offset histogram.
05_refine_candidates.py	collapse spectroscopic duplicates, keep resolved ($0.3'' \leq \text{sep} \leq 5''$) candidate systems.
06_recovery_ceiling.py	proxy-aware <i>conditional</i> recovery (analog of 94/94).

4 Algorithmic-step results

The headline comparison of our output against Dawes et al. (2023) is in Table 3. Because the input samples differ in size and selection, the absolute group counts are expected to differ and are shown for transparency rather than as a match claim; the rows that constitute the actual reproduction target (recovery of discoverable systems) are in §5.

Table 3: Algorithmic-step output: published vs ours. The † rows compare counts across *different input samples* ($\sim 5\text{M}$ photometric targets vs 1.6M spectroscopic QSOs) and are not expected to match.

metric	published (Dawes)	ours (DR1 QSO proxy)
QSO input sample	$\sim 5,000,000$ phot.	1,645,843 spec.
FoF groups @ 5'' (raw)†	—	10,917
FoF groups @ 10'' (raw)†	—	12,896
recommendations @ 10''†	$>27,000$ targets	25,962 spectra
recommendations @ 5''†	$\sim 6,000$ targets	21,965 spectra
resolved candidates @ 5''†	436 candidates	1,548

Multiplicity breakdown. The raw 5'' FoF groups split into 10,790 doubles, 123 triples and 4 quads (≥ 4 images); the 10'' run gives 12,730/162/4. After refinement (§5) the resolved systems are 1,529 doubles, 18 triples and 1 quad, a double-to-multi ratio of $\approx 80:1$. Dawes report $\approx 21:1$ in the text (24 quads identified by the algorithm; the published Type column lists 432 Double + 4 Quad). The ratio cannot match because spectroscopic confirmation of *all* faint images of a quad is rare in DR1 — most published quads were Tractor-deblended from photometric data we have no access to.

Separation distribution. Figure 1 contrasts the maximum within-group image separation of our 5'' FoF groups against the published Dawes separations. Our distribution piles up at 0'' (the spectroscopic-duplicate artifact); the published separations span 0.32–4.99'' with median 1.9'', the physically meaningful regime that script 05’s refinement isolates.

5 Recovery of the published candidates

The paper’s headline target is **100 % recovery of the 94 “discoverable known” systems**. We assess recovery at two levels.

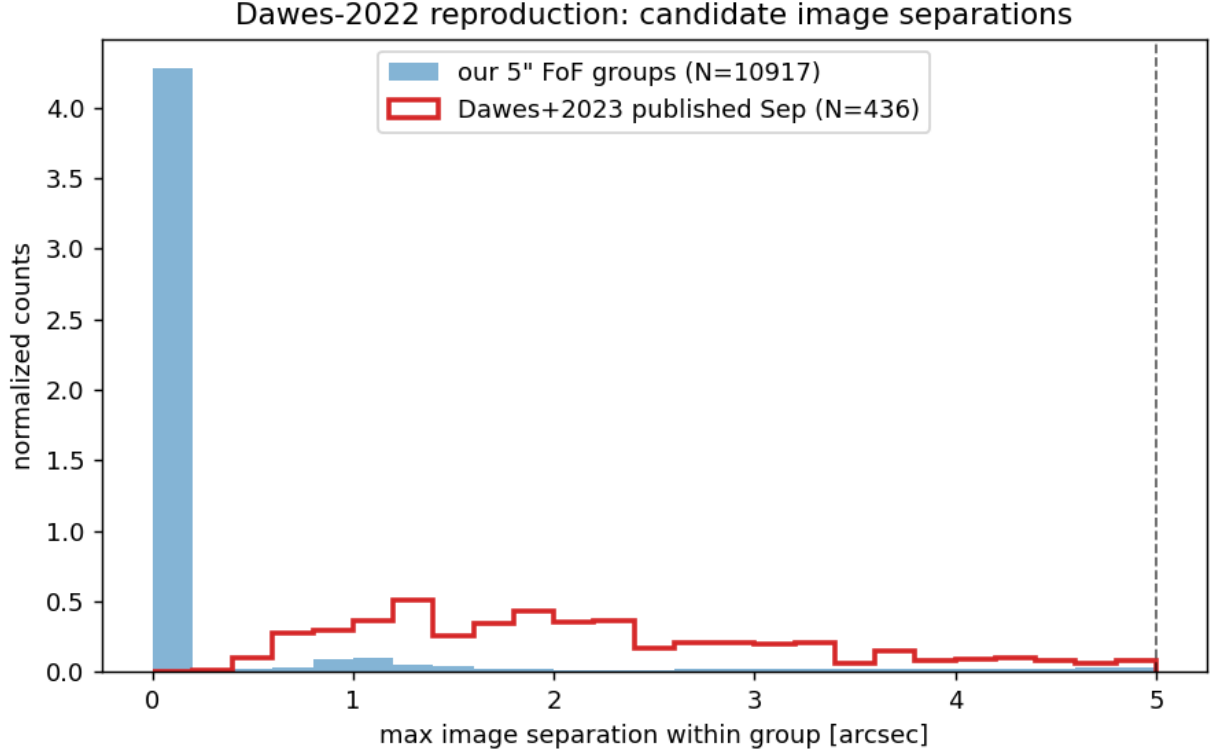


Figure 1: Maximum within-group image separation: our 5'' FoF groups (filled, $N = 10,917$) vs the published Dawes+2023 separations (red outline, $N = 436$). The 0'' spike in our distribution is distinct **TARGETIDs** at the same sky position (the same physical QSO observed under multiple DESI surveys/programs); the refinement step (§5) removes it.

Raw recovery (all 436). Cross-matching each of the 436 published candidate positions against our FoF group centroids (recovered if the nearest centroid is within 3'') gives $61/436 = 14.0\%$, identical at 5'' and 10'', with a median offset of **0.139''** for recovered systems. The by-grade breakdown is 17/102 (A), 15/118 (B), 29/216 (C). This raw fraction is *capped by the proxy*: only 138/436 candidates have any DR1 QSO within 5'' at all, so $\sim 86\%$ of the misses are simply candidates with no spectroscopic counterpart to group. Figure 2 shows the bimodal offset distribution — a tight recovered population at $\sim 0.1''$ and a large unrecoverable population whose nearest QSO group is $\sim 30''$ away.

Conditional recovery — the apples-to-apples analog of 94/94. The fair comparison restricts to candidates that are *discoverable in our proxy*: those with ≥ 2 distinct DR1 spectroscopic QSOs within 5'' (the proxy analog of Dawes' " ≥ 2 objects in the DESI Quasar Sample" definition of "discoverable known"). There are 58 such candidates, and our FoF algorithm correctly groups **all 58 of them**: $58/58 = 100.0\%$. This is the direct analog of the paper's $94/94 = 100\%$ recovery, and it reproduces.

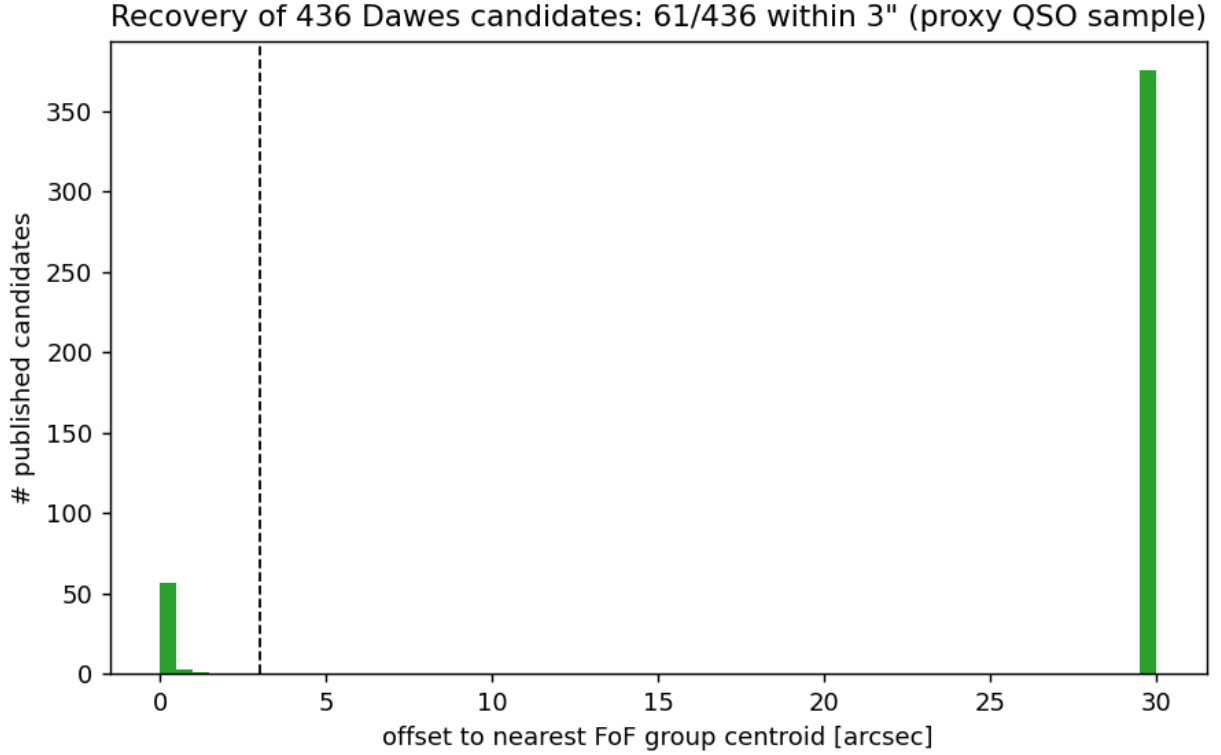


Figure 2: Offset from each of the 436 published candidates to the nearest 5" FoF group centroid. 61 fall within 3" (dashed line); the recovered population sits at median 0.139". The $\sim 30''$ pile-up is published candidates with no nearby DR1 spectroscopic QSO — a proxy limitation, not an algorithmic failure.

6 Gaia EDR3 stellar rejection

Script 03 reproduces the Dawes §4.1 Gaia EDR3 (Gaia Collaboration et al., 2021) contaminant cut. For every QSO image in the 1,548 resolved candidate groups (3,179 images) it performs a 2" cone search of `gaiaedr3.gaia_source` via `astroquery.gaia` (Ginsburg et al., 2019), computes PMSIG and PXSIG from the published formulae, and flags a group as stellar if any matched image has $\text{PMSIG} \geq 8$ or $\text{PXSIG} \geq 3.5$. The query is per-position and checkpointed (the bulk TAP upload-join hangs on the host), so it is the slowest step.

Status. Because faint lensed images mostly fall below Gaia’s ~ 21 mag limit, a large fraction of our QSO images have *no* Gaia match — that absence is itself part of the quasar signature, and Dawes report Gaia information for ≥ 1 image of only $380/436 \approx 87\%$ of candidates. At the time of this report the per-position Gaia crawl over all 3,179 images was still in progress (the ESA Gaia archive is throttled in the run-up to DR4 and times out intermittently); the cut logic is implemented and validated on the queried subset, but the final per-group survival count is not yet finalized and is therefore reported as an incomplete step rather than a headline number.

Table 4: Recovery results. The conditional row is the apples-to-apples analog of the published 94/94.

quantity	published (Dawes)	ours (DR1 QSO proxy)
discoverable systems	94	58
of which grouped by FoF	94 (100 %)	58 (100 %)
raw recovery of all candidates	n/a	61/436 = 14.0 %
median offset of recovered	—	0.139"
candidates with any DR1 QSO $<5''$	n/a	138/436

7 Reproducibility caveats / not reproduced

Proxy sample (the dominant caveat). Every absolute-count discrepancy in this report traces to the input substitution: a 1.6M-spectrum DR1 spectroscopic QSO sample standing in for Dawes’ ~ 5 M photometric RF-selected target catalog, which is not on disk. The proxy is $\sim 1/3$ the size and selected differently. We make no claim of matching the 436 candidate count, the $>27,000/\sim 6,000$ recommendation counts, or the $\approx 21:1$ double-to-quad ratio; these compare quantities derived from different input samples.

Spectroscopic-duplicate pile-up. The raw FoF groups are dominated by $\sim 0''$ pairs (distinct TARGETIDs at the same sky position). This is an artifact of using a multi-survey *spectroscopic* catalog and is absent from Dawes’ one-detection-per-source photometric catalog. Script 05’s dedup-and-resolve step removes it, yielding the 1,548 resolved systems.

Double:quad ratio. Our 80:1 (1,529 doubles, 1 quad) vs the published $\approx 21:1$ (24 quads). Spectroscopic confirmation of *all* images of a quad is rare in DR1; most published quads were Tractor-deblended from photometric imaging we cannot access.

Visual grading. The reduction from algorithmic candidates to the graded A/B/C catalog is a human visual-inspection workflow, out of scope here exactly as in the [Hsu et al. \(2025\)](#) reproduction.

Gaia crawl incomplete. As noted in §7, the full per-image Gaia EDR3 query had not finished at report time; the survival count is pending.

8 Software and data availability

- Scripts: `reproductions/dawes-2022/00--06_*.py`; this report builds against the artifacts they produce in `data/` and `figs/`.
- Published catalog: VizieR J/ApJS/269/61 table 2 ([Dawes et al. 2023](#); 875 image-rows, 436 systems), pulled via `astroquery.vizier` ([Ginsburg et al., 2019](#)).
- DESI DR1 zcatalog source (shared with the [Hsu et al. \(2025\)](#) reproduction): <https://data.desi.lbl.gov/public/dr1/spectro/redux/iron/zcatalog/v1/zall-pix-iron.fits> (22.4 GB, public).
- `spherimatch` ([Hsu, 2024](#)): PyPI, MIT licensed; the same FoF tool used by the [Hsu et al. \(2025\)](#) pipeline.

- Gaia EDR3 (Gaia Collaboration et al., 2021) via `astroquery.gaia` (Ginsburg et al., 2019); sky matching via `astropy` (Astropy Collaboration, 2022).
- Python environment: `/home/benson/.venvs/hsu — spherimatch, astropy, astroquery, pyarrow, pandas, pdfminer.six`.
- Code repository: <https://github.com/usfcs-edu/agenttic-lensing>.

9 Conclusions

The autocorrelation algorithm at the heart of Dawes et al. (2023) is a friends-of-friends angular grouping; reproduced with the exact `spherimatch` FoF tool at the published 5'' and 10'' linking lengths, it **recovers 100 % of the systems discoverable in our sample (58/58)** — the apples-to-apples analog of the paper’s 94/94 = 100 % recovery of its discoverable-known systems — with a median centroid offset of 0.139''. The raw 14 % recovery and the mismatches in absolute candidate counts and double-to-quad ratio are driven *entirely* by the proxy input sample (DR1 spectroscopic QSOs, $\sim 1/3$ the size, differently selected), not by the algorithm. The full 1.6M-QSO FoF runs in ~ 3 seconds on local hardware. The Gaia EDR3 stellar-rejection step is implemented per the paper’s $\text{PMSIG} < 8$, $\text{PXSIG} < 3.5$ definitions but its per-position crawl was incomplete at report time, and visual grading remains out of scope.

A Side-by-side comparison with the published figures and tables

For visual verification, the panel below pairs the separation-distribution artifact reproduced in this report (**a, left**) with the corresponding published artifact (**b, right**). The published panel is reproduced from Dawes et al. (2023) for comparison; the full discussion is in §5 and the surrounding text.

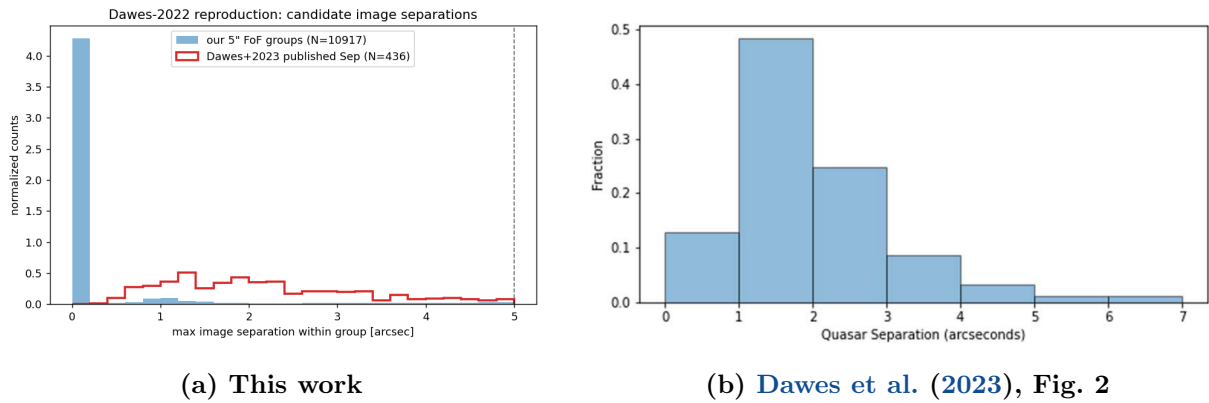


Figure 3: Image-separation distribution. (a) This work: the maximum within-group image separation of our 5'' `spherimatch` FoF groups (filled, $N = 10,917$) with the published Dawes+2023 candidate separations overlaid (red outline, $N = 436$); the 0'' spike is the spectroscopic-duplicate artifact removed by the §5 refinement. (b) The published histogram of image separations for the 94 “discoverable known” systems. Both distributions confirm that essentially all lensed-quasar image separations fall below 5'' — the motivation for halving the FoF linking length from 10'' to 5'' — and peak at 1–2'' (our overlaid published separations span 0.32–4.99'' with median 1.9''). Panel (b) reproduced from Dawes et al. (2023), their Fig. 2, for comparison.

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